MR-Vis: A tool for the **visualisation** of high-dimensional Mendelian randomization results

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Different **measures of increased adiposity** show similar **metabolic profiles** – can we use profiles to explore relationships with disease?

INTRODUCTION

Mendelian randomization analyses predominantly focus on single exposure and single outcome associations. However, **biological pathways involve numerous processes** and **identifying** meaningful **intermediate associations** that can be take forward for further analysis **is complex**.

Metabolite and other omic **effects rarely occur in isolation**. This makes **identifying individual causal associations difficult**. Using **profiles** can **capture** these **combined effects** and help to identify associations for further analysis.

Similarly, **using** an **individual exposure for a complex trait** like increased adiposity **has limited resolution**. Using **multiple** correlated **exposures** each with its own limitations **can increase confidence** in the causal associations. **Combining profiles and multiple exposures** may aid in identifying meaningful associations. Using measures of increased adiposity and metabolites as an example, **we set out to**:

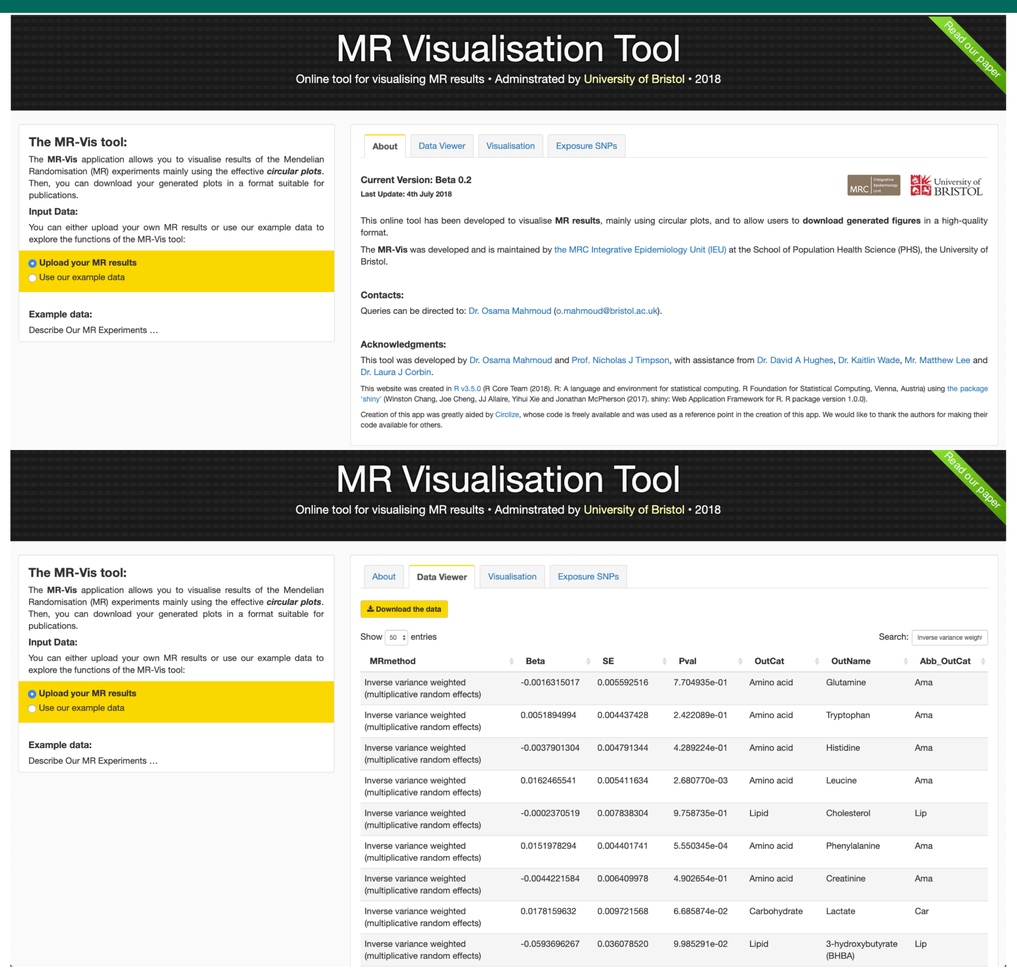
- Build an **application** that **automates** the **visualization** of causal inference analysis
- Allow users to **alter** content and **format** to **identify patterns** in visualizations

MR-Vis FEATURES

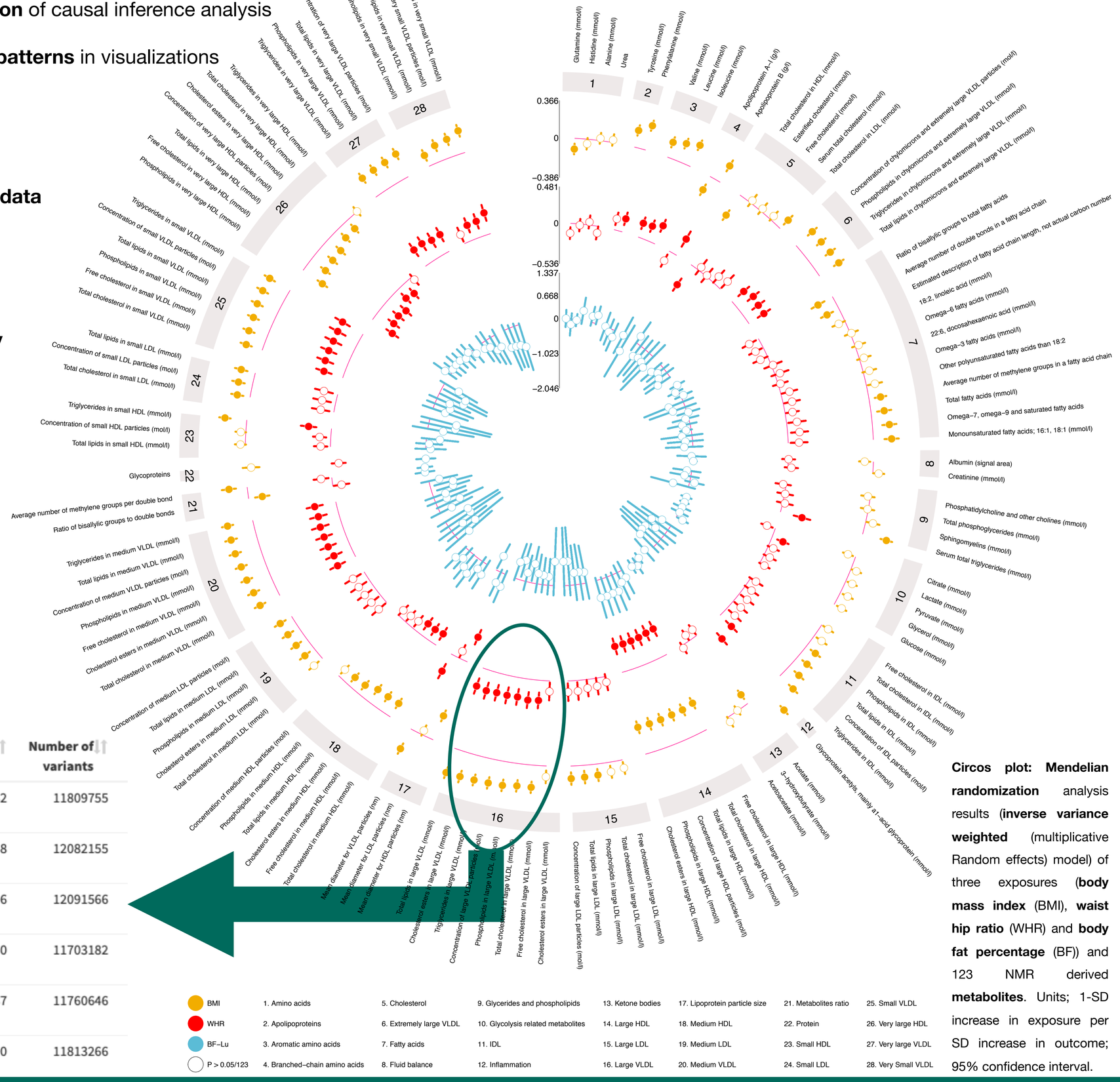
- Users can use **example data** and **upload** their own **data**
- Use **priors** to **alter** the **layout** of the plot
- **Identify profiles** across multiple exposures
- **Compare** exposure-outcome **associations globally**

FUTURE

- **Extract instruments** for selected profile(s) for further applied analysis by integrating with MR-Base
- **Highlight shared SNPs** across all outcomes when selecting profiles to extract instruments
- **R package**



	Trait	Note	First author	Sample size	Number of variants
838	Acetoacetate	AcAce	Kettunen	19262	11809755
839	Acetate	Ace	Kettunen	24748	12082155
840	Alanine		Kettunen	24796	12091566
841	Albumin	Alb	Kettunen	18960	11703182
842	Apolipoprotein A-I	ApoA1	Kettunen	20687	11760646
843	Apolipoprotein B	ApoB	Kettunen	20690	11813266



Circos plot: Mendelian randomization analysis results (**inverse variance weighted** (multiplicative Random effects) model) of three exposures (**body mass index** (BMI), **waist hip ratio** (WHR) and **body fat percentage** (BF)) and 123 NMR derived **metabolites**. Units; 1-SD increase in exposure per SD increase in outcome; 95% confidence interval.

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